

INST 327 (0302)
Team Project Report
12/15/25
Group 8

Lizzie Colvin, Grady Mumm, Philip Kucher, Emmanuel Rhodes, Jordan Stone

Introduction

In this project, our group created a database using cleaned data related to patients with different types of diabetes. This includes information on individuals diagnosed with conditions like Type 1 Diabetes, Gestational Diabetes, MODY, Wolfram Syndrome, and Cystic Fibrosis-Related Diabetes. The goal of this database was to organize this large set of health-related information in a way that makes it easier to store, manage, and analyze. The database was designed with a relational structure to make sure there was consistent data being retrieved.

Our dataset includes a wide range of factors like age, insulin levels, body mass index (BMI), physical activity, dietary habits, blood glucose levels, lab test results, and pregnancy history. To make the database more organized and condensed, we structured it into parts based on patient age, which allowed for easier comparisons throughout a certain age group. We also cleaned and standardized that data to fix inconsistencies and to make sure all information fits well into our database. This process helped reduce errors and ensured that similar data was stored in a consistent format across all tables.

By organizing the data this way, the database makes it easier for anyone using it to look for connections between different factors, join information across many categories, and analyze certain information for the different types of diabetes. Additionally, the database can serve as a resource for understanding how lifestyle, medical history, and age may influence disease outcomes. It shows how structured data design can be and how it supports health decision-making.

Database Description

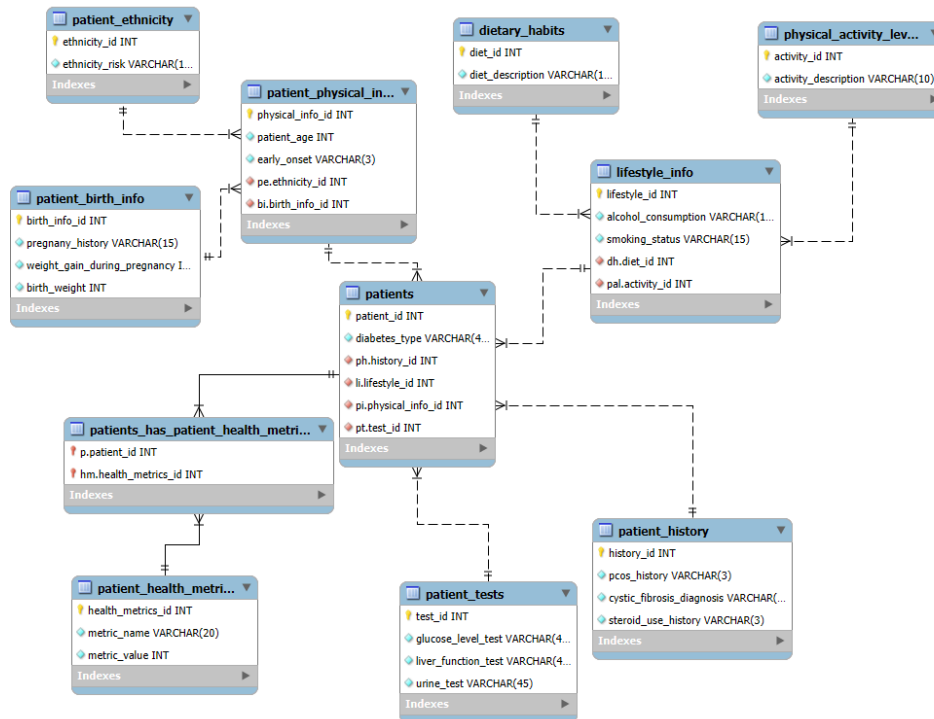
Our main table for our database is the patients' table, which branches out into four many-to-one different tables: patient_physical_info, lifestyle_info, patient_history, and patient_tests, and one one-to-many table: patients_has_patient_health_metrics. Three of these tables also branch out into their own tables, like patient_physical_info linking to patient_ethnicity and patient_birth_info, and lifestyle_info linking to dietary_habits and physical_activity_level, which helps keep the data organized and avoids unnecessary duplication.

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Then patients_has_patient_health_metrics linking to patient_health_metrics. This is also our many-to-many relationship.

Logical Design



For our logical design, we centered the database around the patients table, since it's the main table of all the information we wanted to track. Each patient is connected to their medical history, lifestyle habits, physical details, and test results through foreign keys, which keeps everything organized and makes it easy to follow.

We then separated the patient background and demographics into their own tables. For example, patient_physical_info stores age and early onset indicators of diabetes, and it links to the patient_ethnicity and patient_birth_info to gather the ethnic risk and birth details. Lifestyle information is broken into diet and physical activity tables so we don't have to repeat data and can easily query what we need.

Medical history and test results also have their own tables to keep the health data clean and consistent. For health metrics, we set up a many-to-many relationship with the joining table

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patients_has_patient_health_metrics. This way, patients can have multiple health metrics, and each metric can relate to multiple patients, which matches how health and data really work

Some parts, like diet and activity, could technically be many-to-many, too, but we kept them one-to-many to keep things simpler while still capturing all the important details. Overall, the design keeps data organized, avoids unnecessary repetition, and makes it easy to analyze patterns in patient health.

Physical Database

For our physical database, we retrieved the data from the original diabetes document. We then created all the tables in MySQL Workbench according to our ERD, making sure each table had the correct primary keys, foreign keys, and data types. From there, we focused on organizing the data in a way that made sense for how it would actually be used. Patient information was separated into physical details, lifestyle habits, medical history, and health metrics to keep the data organized and reduce repetition. We also adjusted the structure based on what was available in the original dataset and what would be most useful for analysis. Some things that were inconsistent or not relevant to our research questions were left out, while others were grouped into separate tables to better reflect certain relationships.

Sample Data

Since our original dataset is large, we planned to focus on the parts of it that were the most important to understanding, like weight, health levels, dietary history, and recent or past lifestyles. Some variables that didn't directly impact our dataset were waist circumference, neurological assessments, environmental factors, and a few others, so they were left out to keep things clear and manageable. Then, because the dataset contains about 70,000 records, we used a smaller cleaned amount of data that still included a variety of patients and ensured that the key relationships between variables were kept, so our results stayed useful.

Specifically, we first focused on patients within the age range of eighteen to twenty-five years old. By looking at this age group, we were able to better understand how diabetes has affected those in early adulthood. This reduced the dataset down to around 8,000 rows, but

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further filtering was needed to effectively answer the questions that we created about our data. To do this, we filtered out patients who had made unhealthy lifestyle choices. This includes those with low physical activity levels, unhealthy dietary habits, smokers, those with moderate or high alcohol consumption, and those who have used steroids. This left us with about 246 columns, which was a manageable number of diabetes patients who have made healthy lifestyle choices. Later, we realized we needed to bring this number down, and we decided to filter by low risk of ethnicity, finally bringing ourselves down to 62 rows. Additionally, with regard to sample data, it is important to note that not every table has 15 rows, which, for the sake of our filtering, and the limited variety of the data itself as a result of aforementioned filtering, is to be expected. Some tables only have so many options and variety, making it difficult to reach 15 rows and, as such, they do not have those 15 rows.

Views/Queries

View name	Req. A	Req. B	Req. C	Req. D	Req. E
average_age_by_db_type	X		X		
patients_and_their_health_metrics	X			X	
patients_above_average_bmi	X	X	X		X
test_count_by_diabetes_type	X	X	X		
patient_demographics_lifestyle	X	X			

Query 1 (average_age_by_db_type): This view shows the average age of patients for each diabetes type, helping us see how different types of diabetes relate to age.

Query 2 (patients_and_thier_health_metrics): This view lists each patient along with their recorded health metrics, making it easy to see which measurements are associated with each individual.

Query 3 (patients_above_average_bmi): This view identifies patients whose BMI is higher than the overall average, highlighting individuals who may be at increased health risk.

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Query 4 (test_count_by_diabetes_type): This view displays the number of medical tests recorded for each diabetes type, allowing for easy comparison across groups.

Query 5 (patient_demographics_lifestyle): This view combines demographic and lifestyle information for non-smoking patients to better understand how these factors relate to diabetes.

Changes from the Original Design

When looking at our original table, we noticed that a lot of changes had to be made to get to our current design. Starting off, most of our tables did not have primary keys, so we decided to change this and made sure every table had its own unique primary key id. Afterwards, we made sure that all the foreign keys matched up correctly and linked to the right tables.

Then, moving on to each table individually, our patients' table did not have any foreign keys, mainly because of the primary key issue, and it only had the diabetes_type field, which caused problems since we needed it to connect to multiple tables. We also noticed that the patient_birth_info table could be linked to the patient_physical_info table since birth-related details are part of a patient's physical background rather than standalone patient data, which helped reduce redundancy and improved the overall organization of the database. Our patient_tests table had to be changed from the type and result to glucose level, liver function, and urine test, mainly because we thought this would be easier to work with.

Lastly, we need a many-to-many relationship because our original design only had one-to-many relationships. We did this by getting rid of the patient_health_info table in our original design and creating a patients_has_patient_health_metrics table along with a patient_health_metrics table. This setup allowed each patient to be linked to multiple health metrics and each health metric to be associated with multiple patients. Overall, these changes helped us create a more organized database by adding primary keys, revising foreign keys, and establishing a many-to-many relationship.

Database Ethics Considerations

Diabetes affects people differently across age groups, race, as well as various financial and cultural backgrounds. It was of great importance that our database took these patient

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characteristics into account. The initial set of diabetic patient data from Kaggle does not go into great detail regarding patient demographics beyond basic information. For example, gender isn't a factor in the data, and ethnicity is only represented as either at risk or not. We made an effort to include certain attributes in order to be more inclusive and representative of patients from underrepresented backgrounds. This helped avoid skewness of data and reflected the full diversity of the population that is impacted by diabetes. This also improves healthcare analytics.

Along with implementing more demographic characteristic attributes in the database, it is important that the data itself was collected without any biases or misrepresentations so that researchers don't come to inaccurate conclusions. By ensuring that health measurements were gathered and stored in a consistent way across all groups, the database avoids prioritizing one population over another. In combination with patient data having the ability to be modified at any time to fix potential errors, health research will be both more reliable and inclusive.

Since our database is based on personal health information from Diabetic patients, privacy and fair use of data are very important concerns. Even though the dataset is already anonymous, there needs to be a framework put in place to make sure that patient data is stored properly, protected, and only accessed or modified by those with proper permissions. It is imperative that when handling personal information such as medical history, medication use, and other such data that it follows the proper legal and ethical guidelines.

The purpose of our database is strictly for research by doctors and health professionals, therefore, the data needs to remain confidential so it cannot be misused in any way. It is vital that the information provided is used transparently and responsibly. By being mindful of how the data is secured and who has access to it, our database will remain an ethical resource that respects patient confidentiality and digital privacy.

Lessons Learned

Working on this database taught us a lot about how important careful planning and organization are when building a relational database. One of the biggest takeaways was how important primary keys and foreign keys are for keeping data connected and maintaining integrity. Without them, it's easy for information to become messy and inconsistent.

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We also learned that the value of breaking data into separate tables is to avoid redundancy. For example, separating physical information, lifestyle habits, and medical history made it much easier to manage and query the data. Setting up a many-to-many relationship for health metrics showed us the reality of real-world complexity. This was our hardest part, especially getting the ids and data to match.

Another lesson was the importance of working with cleaned and filtered data. The original dataset had tens of thousands of rows, and trying to use it all would have been overwhelming and redundant. Learning how to filter and select the most relevant information helped us focus on what was important and made the database more practical to use.

Potential Future Work

For potential future work in the diabetes dataset we can explore deeper into other age ranges. Expanding the age range of all patients from the original dataset can allow us and healthcare providers to understand how diabetes affects different age groups, allowing for the data to be more representative and learn how diabetes progresses within the whole patient population. We can create roughly four different datasets of age ranges. The four datasets will involve working with pediatric patients (0-17), young adults (18-39), middle aged (40-64), and seniors (65+). Working with these age groups can help us understand how diabetes presents itself within them and identify which type of diabetes is most common in each range. These discoveries could support new treatments and prevention strategies tailored for each specific group. Additionally, these age group analysis can connect and reveal how other variables are affected as people with diabetes age.

With these datasets we can turn each age range into its own respective database. For each database we can create complex queries that compare and join the different age ranges, helping us find different risk factors within the groups. We can also work on implementing more views to summarize each patient's outcome for each age group and diabetes type, making the data easily accessible and interoperable. This can allow us to examine how these risk factors interact with the age ranges, bringing even more insight on patterns that are not easily seen.

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Citations

Batra, Ankit. "Diabetes Dataset." Accessed 2025. kaggle, 2024,
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